

launched in fire-and-forget mode, destroying targets within the line-of-sight of the launcher and also in fire, observe and update guidance mode while following a top-attack flight trajectory.

In fire-and-forget mode, the tracker is locked on to the target. The missile is launched and it automatically propels itself towards the target. The missile uses a tandem-charged high explosive anti-tank (HEAT) warhead that can penetrate explosive reactive armour.

The guidance system of Spike comprises a charge-coupled device (CCD) and an IIR seeker. The IIR sensor, in addition to providing higher sensitivity, offers improved thermal background rejection characteristics for all-weather day and night operation.

Different variants of Spike missile system include Spike short range (Spike SR) with a maximum opera-

tional range of 800m, Spike medium range (Spike MR) with maximum operational range of 2500m, Spike long range (Spike LR) with maximum operational range of 4000m, Spike extended range (Spike ER) with maximum operational range of 8000m, Spike non line-of-sight (Spike NLOS) with maximum operational range of 25km and Mini Spike with engagement range of 1300m. Mini Spike is an anti-personnel-guided weapon.

The Spike missile system is currently in service with Dutch, Chilean, Colombian, Finnish, German, Polish, Italian, Peruvian, Spanish and Singaporean armed forces. The missile system has been successfully used during Lebanon war in 1982, Second Intifada beginning year 2000, Afghanistan war from 2001 till date and Iraq war in 2006.

Nag (Fig. 15) is a third-generation fire-and-forget ATGM from India, designed and developed by Defence Research and Development Organisation (DRDO) and manufactured by Bharat Dynamics Ltd. It has two variants employing an active IIR seeker and a millimetric wave seeker. The operational range for the land version is 500m to 4000m, and 7km to 10km for air-launched version. It is likely to be inducted into service in 2015.

Surface-to-air (SAM) missiles. Short- to medium-range missiles used in the air defence role against attack helicopters and aircraft are the most common surface-to-air missiles

that use IR homing guidance. A large proportion of these missiles belong to the category of MANPADS. Some of the more common and better known IR-guided surface-to-air missile systems are Stinger, Igla (Igla and Igla-1) and Strela (Strela-2 and Strela-3).

FIM-92 Stinger is a man-portable IR-guided surface-to-air missile designed by General Dynamics and manufactured by Raytheon Missile Systems in the USA, in Germany by European Aeronautic Defence and Space Co. (EADS) and in Turkey by ROCKETSAN. It entered into service in 1981 and continues to be in service till date. It is adaptable to be shouldered fired [Fig. 16(a)] or from land vehicles [Fig. 16(b)] as surface-to-air missile and helicopters as air-to-air missile. Stinger missile has evolved over the years and has undergone significant technological improvements.

Three main variants of stinger include FIM-92A, FIM-92B and FIM-92C. These are known by the names of Stinger basic (FIM-92A), Stinger passive optical seeker technique or Stinger POST (FIM-92B)], and Stinger reprogrammable microprocessor or Stinger RMP (FIM-92C).

Stinger is intended to fulfil short-range air defence (SHORAD) role till 2018. It consists of a Stinger round encased in a launch tube and separates a grip-stock assembly. Stinger basic employs an IR seeker. Stinger POST uses a dual IR and UV seeker, thereby providing higher immunity to countermeasures as compared to Stinger. Stinger RMP is so-called because of its ability to load a new set of software via an ROM chip inserted in the grip at the depot. The missile has maximum effective firing range of 4.8km and maximum speed of 2.2Mach (750m/s).

Stinger made its debut in warfare in 1982 during Falklands war between the UK and Argentina. Subsequently, it was used in the Soviet War in Afghanistan, Angolan civil war, Libyan invasion of Chad, Chechen War, Sri Lankan civil war and Syrian civil war.

Russian Igla is a MANPADS man-

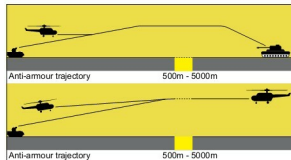


Fig. 14: Flight trajectory in top-attack (top) and direct-attack (bottom) modes



Fig. 15: NAG ATGM



Fig. 16: Stinger missile system being fired (a) from shoulder, and (b) from a land vehicle